

Enterprise Network Intrusion Detection

Official Problem Statement & Operational Scenario

Executive Summary & Context

Your organization operates a complex enterprise network connecting tens of thousands of endpoints and cloud services. The SOC receives hundreds of millions of flow records daily. Your mission is to develop a machine-learning-based Intrusion Detection System capable of identifying malicious traffic with high precision and recall.

Key Constraints & Real-World Challenges

- Asymmetric Operational Risk:** Missing an actual cyber intrusion is vastly more damaging than raising a false alarm. Models must balance decision thresholds appropriately.
- Evolving Adversaries:** Zero-day exploits and unfamiliar attack families will appear in evaluation traffic.
- Dynamic Operating Environments:** Flow distributions shift due to network congestion, latency spikes, and varying link bandwidths.

Hackathon Progression Milestones

- **Hour 1:** Baseline model and initial data exploration.
- **Hour 2:** Advanced feature engineering and algorithm tuning.
- **Hour 3:** Adaptation to asymmetric operational loss function.
- **Hour 4:** Challenge dataset release (unseen threats and shifted traffic).
- **Hour 5:** Final test submission, model artifact submission, and 3-5 min presentation.